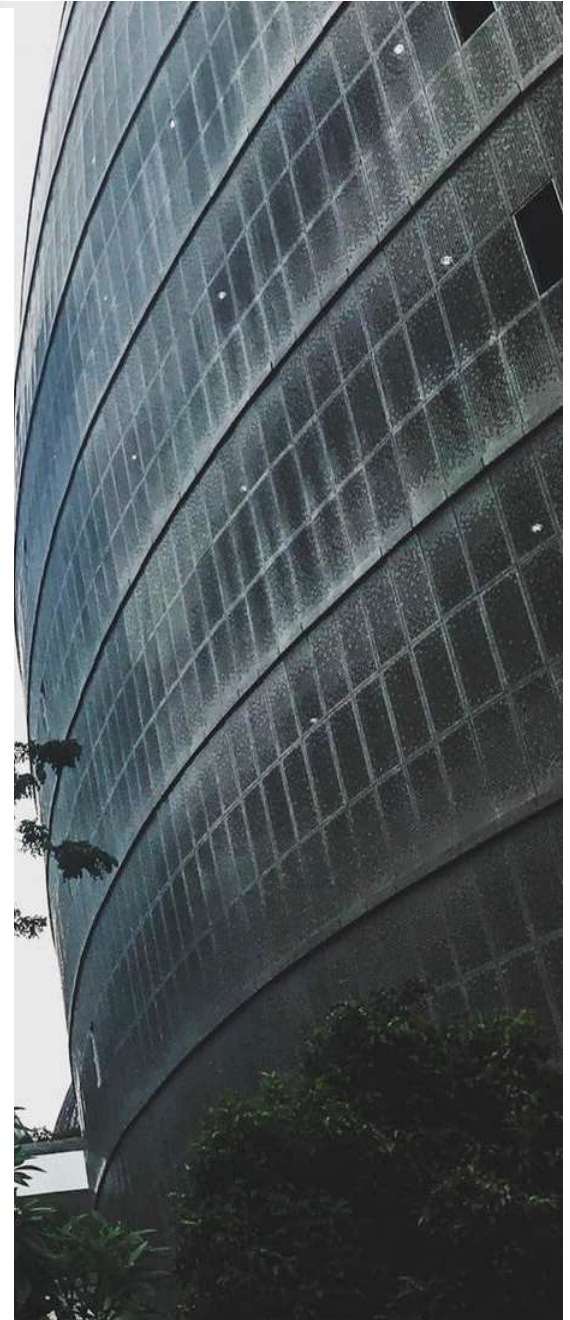


COURSE MATERIAL

**IF670 – Mobile Cross Platform Programming
S1 INFORMATICS GRADUATE PROGRAM
FACULTY OF ENGINEERING AND INFORMATICS**

**INFORMATICS STUDY PROGRAM
FACULTY OF ENGINEERING AND INFORMATION
UNIVERSITAS MULTIMEDIA NUSANTARA**

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MODUL 7 (Animation)

THEME DESCRIPTION

Students will learn how to apply animation on react native.

WEEKLY LEARNING GOAL (SUB-GOAL LEARNING)

Students understand how to add animations to applications in order to enhance user interaction and user experience.

SOFTWARE / HARDWARE REQUIREMENTS

1. Laptop / PC
2. IDE (Android Studio/XCode)
3. Code Editor (Visual Studio Code/Sublime/etc)

PRACTICUM STEPS

1. Create a new Expo project, then update your `index.tsx` file based on the code provided below

```
1  import { StyleSheet, View, Dimensions, Text } from "react-native";
2
3  export default function App() {
4    const screenWidth = Dimensions.get("window").width;
5    const screenHeight = Dimensions.get("window").height;
6
7    return (
8      <View style={styles.container}>
9        <Text>Screen width: {screenWidth}</Text>
10       <Text>Screen height: {screenHeight}</Text>
11     </View>
12   );
13 }
14
15 const styles = StyleSheet.create({
16   container: {
17     flex: 1,
18     alignItems: "center",
19     justifyContent: "center",
20     height: "100%",
21   },
22   box: {
23     height: 120,
24     width: 120,
25     backgroundColor: "#b53df1",
26     borderRadius: 20,
27   },
28 });
29
```

2. Insert the following code into **index.tsx** to determine the screen orientation of your device.

```

4   export default function App() {
5       const screenWidth = Dimensions.get("window").width;
6       const screenHeight = Dimensions.get("window").height;
7
8       //set orientation on your device
9       const [orientation, setOrientation] = useState("potrait");
10      useEffect(() =>{
11          const updateOrientation = () =>{
12              const {width,height} = Dimensions.get("window");
13              if(width<height){
14                  setOrientation("potrait");
15              }else{
16                  setOrientation("landscape");
17              }
18          };
19          Dimensions.addEventListener("change",updateOrientation);
20      });
21  }

```

3. Install the **react-native-reanimated** library in your project by following the guidelines in the official documentation: <https://docs.swmansion.com/react-native-reanimated/>

4. Update your **index.tsx** file to include animations using the **react-native-reanimated** library.

```

23      return (
24          <View style={styles.container}>
25              <Animated.View entering={SlideInLeft}>
26                  <Text>Screen width: {screenWidth}</Text>
27              </Animated.View>
28              <Animated.View entering={SlideInRight}>
29                  <Text>Screen height: {screenHeight}</Text>
30              </Animated.View>
31              <Animated.View entering={SlideInDown}>
32                  <Text>Orientation: {orientation}</Text>
33              </Animated.View>
34          </View>
35      );

```

ASSIGNMENT

Modify your **Week 5 assignment** in **UserList.tsx**. Add animations when rendering the user list. Each user's animation must have a different duration!

For the submission:

1. Create a Git repository with the name **Week7_[NIM]_[Name]**.
2. Push your project files to the repository.
3. Copy the repository link and paste it into **Week7_[NIM]_[Name].txt**
4. Submit to elearning in .txt or .zip format

REFERENSI

<https://docs.swmansion.com/react-native-reanimated/>